

OFFICIAL SDLC SPECIFICATION

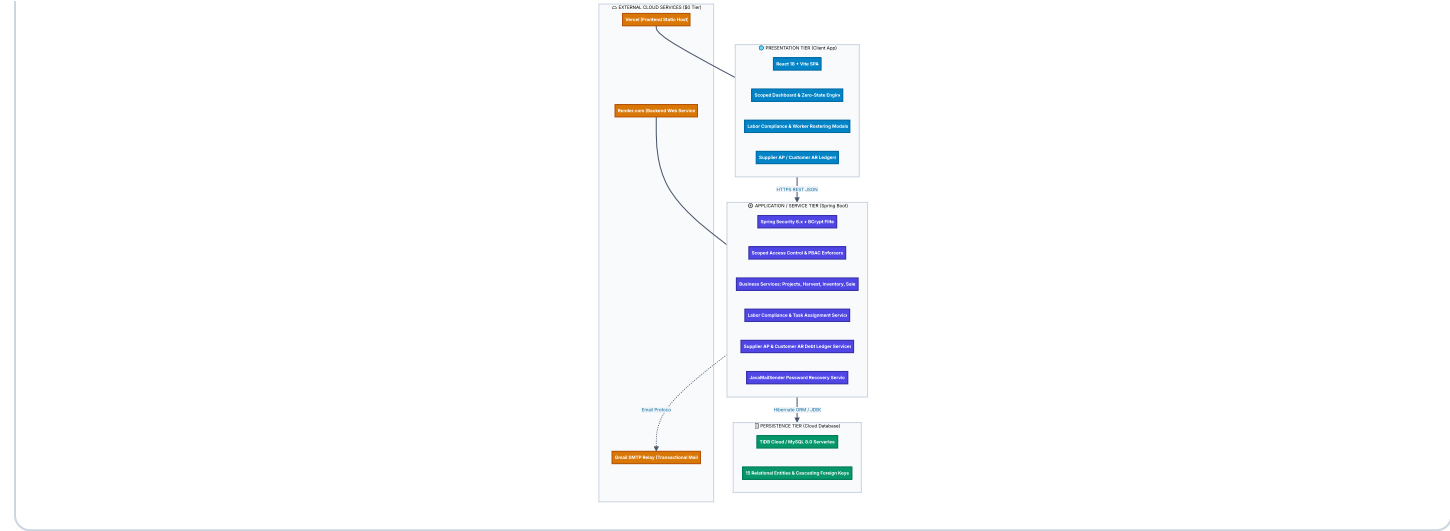
System Design & Architecture Document (SDD)

SmartFarm Management Application

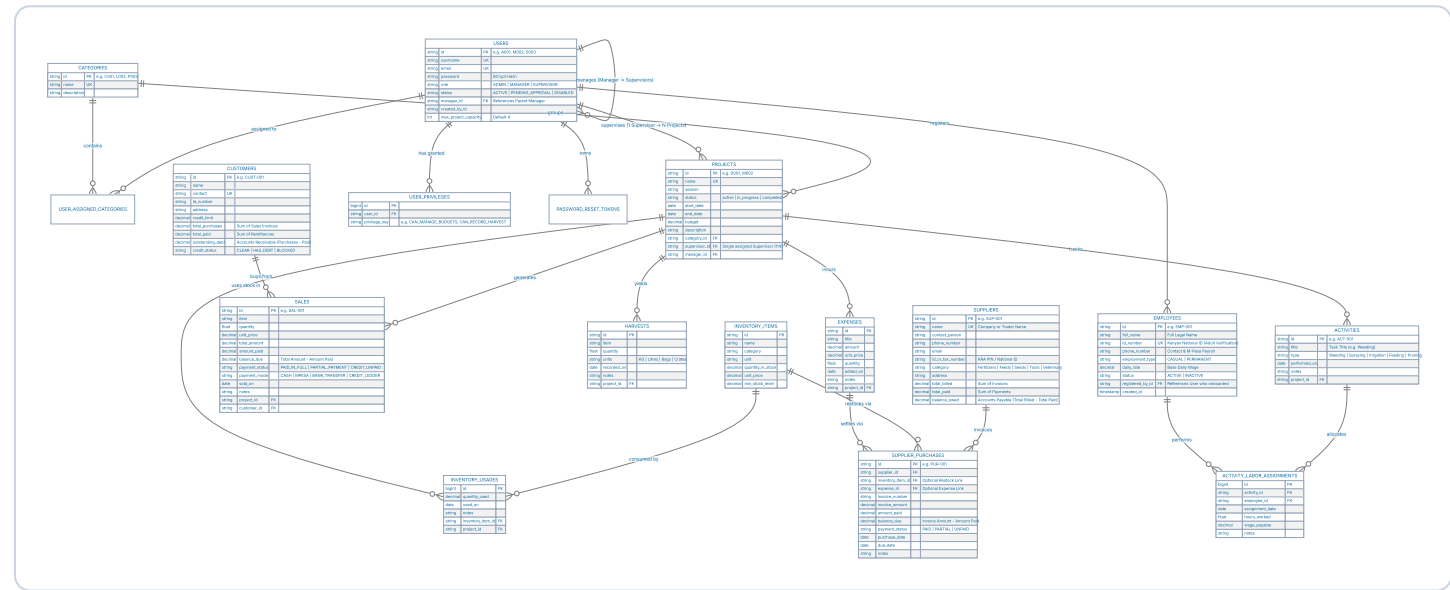
Document Version: 1.0	SDLC Stage: Phase 2 — System Architecture, ERD & Module Design	Status: Approved Technical Blueprint
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1. Architectural Overview & Technology Stack

The SmartFarm system follows a **3-Tier Clean Architecture** separating presentation, business orchestration, and persistent storage:

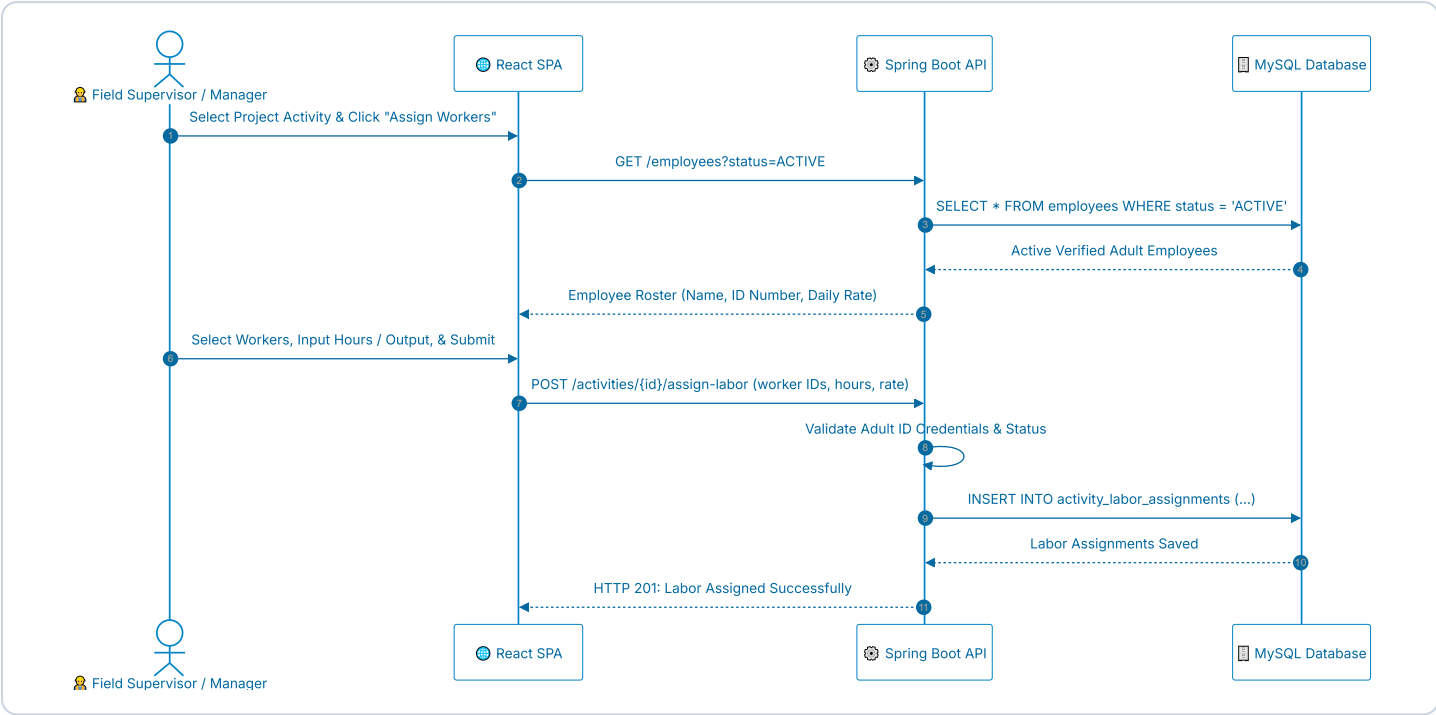


2. Complete Entity-Relationship Model (ERD)



3. Module Breakdown & API Specifications

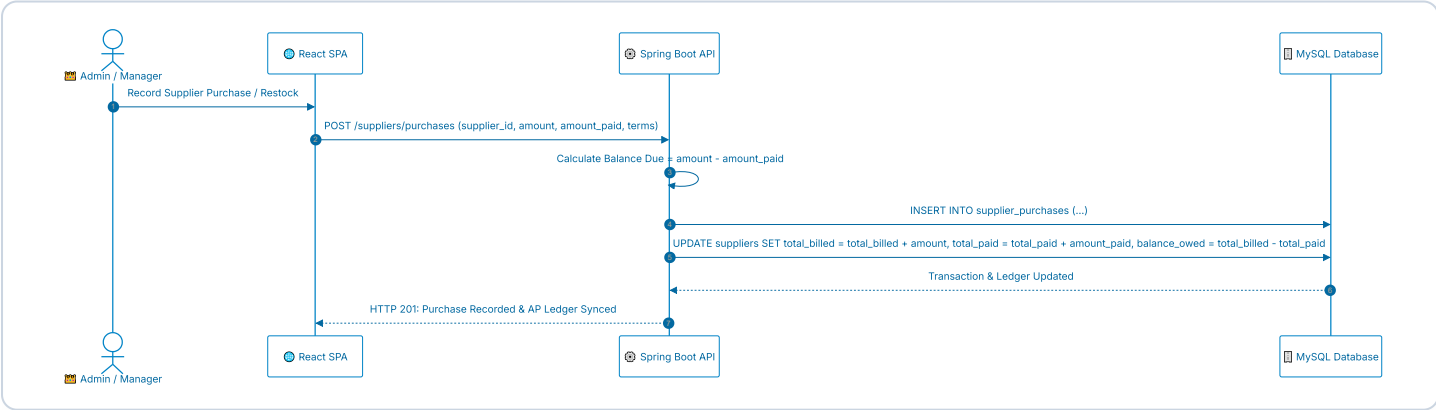
3.1 Labor Compliance & Worker Task Allocation Module



API Endpoints:

- GET /employees — Fetch verified employees list (supports filtering by employment_type , status).
- POST /employees — Register new verified employee (Validates National ID uniqueness & adult status).
- PUT /employees/{id} — Update employee details / status (ACTIVE / INACTIVE).
- POST /activities/{activityId}/assign-labor — Assign verified employees to a project activity.
- GET /activities/{activityId}/labor — Retrieve labor log for an activity with wage totals.

3.2 Supplier Management & Accounts Payable (AP) Ledger



API Endpoints:

- GET /suppliers — Fetch suppliers with live Accounts Payable debt balances.
- POST /suppliers — Register a new supplier vendor.
- PUT /suppliers/{id} — Edit supplier details.
- POST /suppliers/purchases — Record supply invoice and cash/credit split.
- GET /suppliers/{id}/statement — Comprehensive supplier account statement & invoice history.
- POST /suppliers/{id}/payments — Remit debt settlement payment to supplier.

3.3 Customer Credit & Accounts Receivable (AR) Ledger

API Endpoints:

- GET /customers — Fetch customers with credit balances and debt flags.
- POST /customers — Register or update customer details.
- POST /sales/create — Record sale supporting cash, M-Pesa, or credit with amount_paid vs balance_due .
- GET /customers/{id}/debt-history — Retrieve customer outstanding invoices and debt aging.
- POST /customers/{id}/settle-debt — Record payment remittance against customer debt balance.

4. Privilege & Access Matrix (PBAC)

Privilege Key	Description	Admin	Manager	Supervisor
`CAN_CREATE_CATEGORIES`	Create/edit farm categories & sectors	✔	Toggle	✗
`CAN_CREATE_SUPERVISORS`	Register & provision field supervisors	✔	Toggle	✗
`CAN_VIEW_FINANCIALS`	View revenue, cash flow, & AP/AR totals	✔	Toggle	✗ (Shielded)
`CAN_MANAGE_BUDGETS`	Create projects, edit details & budgets	✔	Toggle	✗
`CAN_DELETE_INVENTORY`	Permanently delete inventory catalog items	✔	Toggle	✗
`CAN_MANAGE_SUPPLIERS`	Register suppliers & record AP purchases	✔	Toggle	✗
`CAN_RECORD_HARVEST`	Log crop/animal harvest production yields	✔	✔	Toggle
`CAN_LOG_ACTIVITIES`	Record daily field activities	✔	✔	Toggle
`CAN_USE_INVENTORY`	Deduct supplies & fertilizers for projects	✔	✔	Toggle
`CAN_ASSIGN_LABOR`	Assign verified employees to field tasks	✔	✔	Toggle
`CAN_RECORD_EXPENSES`	Record petty field cash expenses	✔	✔	Toggle
`CAN_RECORD_SALES`	Log direct farm-gate sales (isolated AR)	✔	✔	Toggle